

K. J. Somaiya College of Engineering, Mumbai-77

End Semester Exam

December 2020

Max. Marks: 30

Duration: 1 Hr. 15 Min. (For attempting questions) + 15 min. (uploading)

Class: LY

Name of the Course: Soft computing (Elective)

Course Code: UCEE702

Semester: VII

Branch: COMP

Instructions:

- (1) **All questions are compulsory**
- (2) **Draw neat diagrams**
- (3) **Assume suitable data if necessary**

Question No.		Max. Marks																		
Q.1	A. List and explain in brief different types of activation functions. B. Using the Hebb rule, find the weights required to perform the following classification of the given input pattern shown in the figure. The pattern is shown as a 3X 3 matrix form. The plus symbol represents the value 1 and empty squares indicate -1. Consider 'I' belongs to the member of the class, so the target value is 1 and 'O' does not belong to the member of the class that means it has target value -1. Compute the updated weights <u>only for first input pattern.</u> <table><tr><td></td><td>+</td><td>+</td></tr><tr><td></td><td></td><td>+</td></tr><tr><td>+</td><td>+</td><td></td></tr></table> Pattern 'I' <table><tr><td>+</td><td></td><td>+</td></tr><tr><td>+</td><td></td><td>+</td></tr><tr><td>+</td><td></td><td>+</td></tr></table> Pattern 'O' OR B. Construct a Kohonen's self-organizing feature map to cluster the four given vectors, [1 0 1 1], [1 0 0 1], [0 0 1 0], [0 1 0 1]. The numbers of clusters to be formed are 2. Assume initial learning rate as 0.5. Compute the updated weight matrix <u>only for first input pattern.</u>		+	+			+	+	+		+		+	+		+	+		+	5+5
	+	+																		
		+																		
+	+																			
+		+																		
+		+																		
+		+																		

Q.2	A. Implement OR function with bipolar inputs and targets using Adaline network. Assume and clearly specify the necessary data. Compute the answer <u>only for one epoch.</u> OR A. Train a hetero-associative network to store the input vectors $s = (s_1, s_2, s_3, s_4)$ to the output vectors $t = (t_1, t_2)$. The vector pairs are as per given in the following table. <table border="1"><thead><tr><th>Inputs and targets</th><th>s1</th><th>s2</th><th>s3</th><th>s4</th><th>t1</th><th>t2</th></tr></thead><tbody><tr><td>1st</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>2nd</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>3rd</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>4th</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td></tr></tbody></table> B. Construct an LVQ with five vectors assigned to two classes. The given vectors are as given in following table. Solve the problem <u>only up to the second input vector.</u> <table border="1"><thead><tr><th colspan="4">Vector</th><th>Class</th></tr></thead><tbody><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>2</td></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td><td>2</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td></tr></tbody></table>	Inputs and targets	s1	s2	s3	s4	t1	t2	1st	1	0	0	1	0	1	2nd	1	0	0	0	0	1	3rd	0	1	0	1	1	0	4th	0	0	1	1	1	0	Vector				Class	0	0	1	1	1	1	0	0	0	2	0	0	0	1	2	1	1	0	0	1	0	1	1	0	1	5+5
Inputs and targets	s1	s2	s3	s4	t1	t2																																																													
1st	1	0	0	1	0	1																																																													
2nd	1	0	0	0	0	1																																																													
3rd	0	1	0	1	1	0																																																													
4th	0	0	1	1	1	0																																																													
Vector				Class																																																															
0	0	1	1	1																																																															
1	0	0	0	2																																																															
0	0	0	1	2																																																															
1	1	0	0	1																																																															
0	1	1	0	1																																																															
Q.3	A. Consider a problem of finding comfortable time to travel by Mumbai local train where there is no compulsion to arrive at the destination before any particular time. i. Define the linguistic variable for this problem. (1) ii. Design a fuzzy membership set (2) iii. Suggest the most appropriate Fuzzification method for this problem. (1) iv. Apply the suggested Fuzzification method and plot the graph for the chosen membership function. (2) B. Explain the process of genetic algorithms with a suitable example. OR B. What are CNN? Give any two applications of CNN in two different areas. Explain the same in brief.	5+5																																																																	